PLANT BIOLOGY & ENERGY SYSTEMS

A comprehensive study document for semantic search, RAG and retrieval evaluation

13 chapters • 30 review questions • Glossary • Answer key

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For retrieval evaluation, ask both direct questions and paraphrased questions. Also include questions that combine concepts from multiple chapters, and unrelated questions that should not retrieve relevant passages.

Contents

1. Plant Cells: The Structural Foundation
2. Photosynthesis: Capturing Light Energy
3. The Light-Dependent Reactions
4. The Calvin Cycle and Carbon Fixation
5. Environmental Factors and Photosynthetic Rate
6. Plant Respiration and Energy Use
7. Water, Minerals and Sugar Transport
8. Adaptations to Water Availability
9. Photosynthesis in Ecosystems
10. Experimental Methods and Data Interpretation
11. Worked Examples and Concept Connections
12. Glossary
13. Review Questions for Retrieval and Assessment

1. Plant Cells: The Structural Foundation

1.1 Cell organization

Plants are multicellular eukaryotes. Their cells contain a nucleus, mitochondria, endoplasmic reticulum, Golgi apparatus, ribosomes, a plasma membrane and a cytoplasm. Plant cells also typically contain a cellulose-rich cell wall, a large central vacuole and plastids. The cell wall provides mechanical support and helps maintain cell shape, while the plasma membrane controls the movement of substances into and out of the cell.

The central vacuole stores water, ions, pigments and other dissolved substances. When water enters the vacuole, it presses the cytoplasm against the cell wall and contributes to turgor pressure. Turgor helps keep non-woody stems and leaves firm. If a plant cell loses too much water, the vacuole shrinks and the cell may become flaccid. In severe water loss, the plasma membrane can pull away from the cell wall, a condition called plasmolysis.

1.2 Chloroplast structure

Chloroplasts are specialized plastids in which photosynthesis takes place. A chloroplast is surrounded by an outer and an inner membrane. Inside is a fluid-filled region called the stroma. Suspended in the stroma is a system of flattened membrane sacs called thylakoids. Thylakoids are often stacked into structures called grana; a single stack is a granum. Lamellae connect some of the stacks.

Chlorophyll and other photosynthetic pigments are embedded in the thylakoid membranes as parts of pigment-protein complexes. The thylakoid membrane also contains photosystems, electron carriers and ATP synthase. The stroma contains enzymes used in carbon fixation, including the enzyme RuBisCO. This separation of structures allows the light-dependent reactions and the Calvin cycle to occur in different chloroplast compartments.

1.3 Pigments and visible color

Chlorophyll a is the central photosynthetic pigment in oxygen-producing photosynthesis. Chlorophyll b and carotenoids act as accessory pigments in many plants. Chlorophyll absorbs strongly in blue-violet and red regions of visible light, while green wavelengths are less strongly absorbed and are reflected or transmitted. This is why many leaves appear green.

Carotenoids absorb light in parts of the blue and blue-green range and can transfer some captured energy to chlorophyll. They also help protect the photosynthetic apparatus from damage caused by excess light. Pigment composition can vary among species and can change with leaf age, season and environmental conditions.

2. Photosynthesis: Capturing Light Energy

2.1 Overall process

Photosynthesis converts light energy into chemical energy stored in organic molecules. In oxygenic photosynthesis, plants, algae and some bacteria use water as an electron source and release oxygen. In plants, the process occurs mainly in chloroplasts, especially in green tissues exposed to light.

The simplified overall equation is: 6 CO2 + 6 H2O + light energy -> C6H12O6 + 6 O2. This equation summarizes the net inputs and outputs, but it hides many intermediate reactions. The carbon atoms in the carbohydrate originate from carbon dioxide. The oxygen released as O2 originates from water molecules split during the light-dependent reactions.

2.2 Two coordinated stages

Photosynthesis is commonly described as two connected stages. The light-dependent reactions take place in the thylakoid membranes. They capture light energy, split water, release oxygen, move electrons through an electron transport chain and produce ATP and NADPH. ATP is an energy-transfer molecule; NADPH carries high-energy electrons and hydrogen.

The Calvin cycle takes place in the stroma. It uses ATP and NADPH from the light-dependent reactions to incorporate carbon dioxide into organic molecules. The cycle produces a three-carbon carbohydrate precursor called glyceraldehyde-3-phosphate, often abbreviated G3P. Some G3P can be used to build sugars and other organic compounds, while the rest helps regenerate the carbon-accepting molecule RuBP.

2.3 What happens to the products?

Plants use photosynthetic carbon to build many substances. Glucose can be used in cellular respiration, converted into sucrose for transport, or linked into starch for storage. Carbon also enters cellulose, lipids, amino acids and other molecules. Photosynthesis therefore supports plant growth and supplies much of the chemical energy that moves through food webs.

Not all photosynthetic products remain in the leaf where they are produced. Sucrose is a major transport sugar in many plants. It can move through the phloem from source tissues, such as mature photosynthesizing leaves, to sink tissues, such as roots, fruits, seeds and growing shoots.

3. The Light-Dependent Reactions

3.1 Photosystem II and water splitting

Light-dependent reactions begin when pigment molecules in photosystem II absorb photons. Excited electrons are transferred to an electron acceptor. The electrons lost from photosystem II are replaced by electrons obtained from water. A protein complex associated with photosystem II catalyzes the splitting of water, producing electrons, hydrogen ions and molecular oxygen.

Water splitting is important for two reasons. It supplies replacement electrons to the photosynthetic electron transport chain, and it is the source of the oxygen released by oxygenic photosynthesis. Oxygen diffuses away from the photosynthetic tissue, while protons contribute to the electrochemical gradient across the thylakoid membrane.

3.2 Electron transport and proton gradient

Electrons move from photosystem II through a series of carriers. As electrons pass through parts of the chain, energy is used to move protons into the thylakoid lumen. The concentration of protons becomes higher inside the lumen than in the stroma. This difference in proton concentration and electrical charge is an electrochemical gradient.

Protons flow back toward the stroma through ATP synthase, a membrane protein complex. The energy of this flow drives the formation of ATP from ADP and inorganic phosphate. The use of a proton gradient to power ATP synthesis is called chemiosmosis. Chemiosmosis also occurs in mitochondria during cellular respiration, although the membranes and electron sources differ.

3.3 Photosystem I and NADPH

After leaving the earlier electron carriers, electrons reach photosystem I. Light absorbed by photosystem I raises the electrons to a higher energy state. The electrons are transferred to ferredoxin and then used by the enzyme NADP+ reductase to reduce NADP+ to NADPH.

The light-dependent reactions therefore provide ATP and NADPH to the Calvin cycle. The relative amounts of ATP and NADPH required by carbon fixation can be balanced through changes in electron flow. In cyclic electron flow around photosystem I, electrons return to the electron transport chain instead of reducing NADP+. This pathway can increase ATP production without producing additional NADPH or oxygen.

4. The Calvin Cycle and Carbon Fixation

4.1 Carbon fixation

The Calvin cycle begins when carbon dioxide is attached to ribulose-1,5-bisphosphate (RuBP), a five-carbon molecule. The enzyme RuBisCO catalyzes this reaction. The unstable six-carbon intermediate quickly splits into two molecules of 3-phosphoglycerate (3-PGA), each containing three carbon atoms.

Carbon fixation is the process by which inorganic carbon from CO2 becomes part of an organic molecule. RuBisCO is abundant in many leaves because it performs a central role in carbon entry into the Calvin cycle. However, RuBisCO can also react with oxygen, a reaction that contributes to photorespiration.

4.2 Reduction and regeneration

ATP and NADPH are used to convert 3-PGA into G3P. ATP supplies energy and NADPH supplies reducing power. For every three carbon dioxide molecules fixed, the cycle produces six G3P molecules during the reduction phase. One G3P can leave the cycle as a net product; the remaining five are rearranged, using ATP, to regenerate three molecules of RuBP.

The cycle does not directly produce a six-carbon glucose molecule in one turn. Instead, G3P is the net carbohydrate product. Two net G3P molecules can contribute the carbon needed to form one six-carbon sugar. The Calvin cycle depends on products from the light-dependent reactions, so it is indirectly dependent on light even though its carbon-fixation reactions do not use photons directly.

4.3 Photorespiration

RuBisCO has two competing activities. It can add carbon dioxide to RuBP, supporting the Calvin cycle, or it can add oxygen to RuBP, initiating photorespiration. Photorespiration becomes more likely when oxygen is relatively abundant and carbon dioxide is limited near RuBisCO, a situation that can occur when stomata close during hot or dry conditions.

Photorespiration consumes energy and releases some previously fixed carbon dioxide. It reduces the net efficiency of carbon fixation under certain conditions. C4 and CAM plants have adaptations that concentrate carbon dioxide around RuBisCO or separate carbon fixation from the Calvin cycle in time or space, reducing the impact of photorespiration.

5. Environmental Factors and Photosynthetic Rate

5.1 Light intensity and wavelength

At low light intensity, increasing the amount of light generally increases the rate of photosynthesis because more photons are available to drive the light-dependent reactions. At higher light intensities, the rate may reach a plateau because another factor, such as carbon dioxide concentration or enzyme activity, becomes limiting. Extremely intense light can damage photosynthetic components.

Light quality also matters. Photosynthetic pigments absorb some wavelengths more effectively than others. Blue and red light are generally effective for photosynthesis in green plants, while green light is less strongly absorbed by chlorophyll. However, green light can penetrate deeper into leaf tissues and may still contribute to photosynthesis.

5.2 Carbon dioxide concentration

Carbon dioxide is a substrate for the Calvin cycle. When carbon dioxide concentration is low, increasing it can raise the rate of carbon fixation, provided that light and temperature are not limiting. At sufficiently high carbon dioxide concentrations, the response levels off because another process limits the overall rate.

Stomata are adjustable pores, usually found in the epidermis of leaves. Guard cells regulate their opening. Open stomata allow carbon dioxide to enter, but also permit water vapor to escape. Plants therefore balance carbon dioxide uptake with water conservation. During drought, stomata often close, reducing water loss but also limiting carbon dioxide entry.

5.3 Temperature

Photosynthetic reactions are influenced by temperature because many steps are enzyme-catalyzed. As temperature rises from a low level toward an optimum, enzyme activity and reaction rates often increase. Above the optimum, photosynthesis may decline because enzymes lose function, membranes are affected, respiration increases or stomata close and restrict carbon dioxide uptake.

The optimum temperature varies among species and depends on their environment and acclimation. A plant adapted to a cool climate may have a lower photosynthetic temperature optimum than a plant adapted to a hot climate. Temperature effects should therefore be interpreted in relation to species and growth conditions.

5.4 Limiting factors

A limiting factor is a variable that restricts the rate of a process when it is in shortest supply relative to demand. For photosynthesis, common limiting factors include light intensity, carbon dioxide concentration and temperature. Water availability, nutrient supply, leaf age and stomatal behavior can also influence the rate.

The law of limiting factors is useful for interpreting experiments. If increasing light intensity does not increase photosynthesis, light may no longer be the limiting factor. Carbon dioxide concentration or temperature may then be limiting. Changing one variable at a time while holding others constant helps identify which factor controls the measured rate.

6. Plant Respiration and Energy Use

6.1 Cellular respiration

Plants carry out cellular respiration to release usable energy from organic molecules. In aerobic respiration, glucose is broken down through glycolysis, pyruvate oxidation, the citric acid cycle and oxidative phosphorylation. Most ATP in aerobic respiration is produced through oxidative phosphorylation in mitochondria.

Photosynthesis and respiration are related but distinct. Photosynthesis stores energy in organic molecules and uses carbon dioxide and water as inputs. Aerobic respiration releases energy from organic molecules and uses oxygen, producing carbon dioxide and water. Photosynthetic cells also respire, including during the day; respiration does not simply stop when photosynthesis occurs.

6.2 Comparing chloroplasts and mitochondria

Chloroplasts and mitochondria both contain internal membranes, electron transport chains and ATP synthase. In chloroplasts, light energy drives electron flow and creates a proton gradient across the thylakoid membrane. In mitochondria, energy released from oxidation of food molecules drives electron transport across the inner mitochondrial membrane.

The organelles have different primary roles. Chloroplasts capture light energy and use it to build organic molecules. Mitochondria transfer energy from organic molecules into ATP that can power cellular work. Both organelles contain their own DNA and ribosomes, a feature that is consistent with the endosymbiotic explanation for their evolutionary origins.

6.3 Carbon balance

A plant's net carbon gain depends on the balance between carbon fixed by photosynthesis and carbon released through respiration and other processes. A plant can photosynthesize during daylight and respire continuously. Over a full day, net carbon gain is positive when carbon fixation exceeds carbon loss.

Growth depends on more than the instantaneous photosynthetic rate. Carbon must be allocated to leaves, stems, roots, flowers, fruits and storage tissues. Environmental stress can alter both carbon gain and allocation. For example, drought can reduce photosynthesis through stomatal closure while increasing the need to allocate resources to roots or protective compounds.

7. Water, Minerals and Sugar Transport

7.1 Xylem transport

Xylem transports water and dissolved mineral ions mainly from roots toward stems and leaves. Much of the upward movement in tall plants is explained by the cohesion-tension mechanism. Water evaporates from moist cell walls inside leaves and diffuses out through stomata. This transpiration creates tension that pulls a continuous column of water upward through xylem vessels and tracheids.

Water molecules cohere to one another through hydrogen bonding and adhere to the hydrophilic walls of xylem conduits. These properties help maintain the water column. Root pressure can contribute to water movement in some conditions, but transpiration pull is the major mechanism for water transport in many plants.

7.2 Phloem transport

Phloem transports sugars and other organic substances from sources to sinks. A source is a region that exports sugars, such as a mature leaf producing more sugar than it uses. A sink is a region that imports or stores sugars, such as a developing fruit, root, seed or growing shoot.

The pressure-flow model explains long-distance movement in phloem. Sugars are loaded into sieve tube elements, often with the help of companion cells. Water enters from nearby xylem by osmosis, raising hydrostatic pressure. At a sink, sugars are unloaded and water may leave, creating a pressure difference that drives bulk flow through the sieve tube.

7.3 Mineral nutrition

Plants require mineral nutrients for growth and metabolism. Nitrogen is needed for amino acids, proteins and nucleic acids. Magnesium is part of the chlorophyll molecule and is important for photosynthetic function. Phosphorus is present in ATP, nucleic acids and phospholipids. Potassium contributes to osmotic regulation and enzyme activation.

Deficiencies can produce characteristic symptoms, although symptoms may overlap. Nitrogen deficiency often causes older leaves to become pale or yellow because nitrogen is mobile within the plant. Magnesium deficiency can also cause chlorosis, often beginning in older leaves. A nutrient diagnosis should consider soil conditions, plant species, root health and other possible stresses.

8. Adaptations to Water Availability

8.1 C3 plants

C3 photosynthesis is the most widespread carbon-fixation pathway. RuBisCO fixes carbon dioxide directly into a three-carbon product, 3-PGA. C3 plants often perform well under moderate temperatures and adequate water. In hot, dry conditions, stomatal closure can lower internal carbon dioxide concentration and increase photorespiration.

8.2 C4 plants

C4 plants initially fix carbon dioxide into a four-carbon compound using the enzyme PEP carboxylase. This initial fixation commonly occurs in mesophyll cells. The four-carbon compound is transported to bundle-sheath cells, where carbon dioxide is released and concentrated around RuBisCO. The Calvin cycle then operates in the bundle-sheath cells.

This spatial separation reduces photorespiration and can improve performance in hot, bright environments. C4 photosynthesis requires additional energy compared with the C3 pathway, so its advantage depends on environmental conditions. Maize and sugarcane are familiar examples of C4 plants.

8.3 CAM plants

Crassulacean acid metabolism (CAM) is an adaptation found in many plants from arid environments. CAM plants typically open their stomata at night, when temperatures are cooler and water loss is lower. They take in carbon dioxide and store it in organic acids. During the day, stomata are usually closed and carbon dioxide is released internally for the Calvin cycle.

CAM separates initial carbon dioxide uptake and the Calvin cycle in time rather than in different cell types. Cacti, many succulents and some orchids use CAM. CAM can conserve water, but growth may be slow because carbon dioxide uptake is limited by the amount that can be stored overnight.

8.4 Comparing C3, C4 and CAM

C3 plants fix carbon dioxide directly through RuBisCO in mesophyll cells. C4 plants use a four-carbon intermediate and spatially concentrate carbon dioxide in bundle-sheath cells. CAM plants take up carbon dioxide mainly at night and use stored carbon dioxide during the day. These pathways are adaptations to different combinations of temperature, light and water availability rather than a simple ranking from best to worst.

9. Photosynthesis in Ecosystems

9.1 Primary production

Primary production is the formation of organic matter by autotrophs. Gross primary production (GPP) is the total rate at which producers capture energy and fix carbon. Some of this energy is used by producers for cellular respiration. Net primary production (NPP) is the energy or biomass remaining after producer respiration: NPP = GPP - respiration.

NPP represents the energy available for plant growth, reproduction and transfer to consumers and decomposers. It varies among ecosystems. Temperature, water availability, nutrient supply, growing season length and light all influence primary production. Tropical forests can have high annual productivity, while deserts and some polar ecosystems have low productivity because environmental conditions constrain plant growth.

9.2 Food webs and carbon movement

Photosynthesis introduces energy-rich organic carbon into most terrestrial food webs. Herbivores obtain carbon by eating plants, and carnivores obtain it by eating other consumers. Decomposers break down dead organisms and waste, returning nutrients to the environment and releasing carbon dioxide through respiration.

Energy flows through ecosystems and is eventually dissipated as heat, while matter such as carbon is recycled. Carbon moves among the atmosphere, organisms, soils, oceans and geological reservoirs. Photosynthesis removes carbon dioxide from the atmosphere, while respiration, decomposition and combustion release it.

9.3 Seasonal and daily patterns

Photosynthetic activity changes over the day as light intensity changes. It also changes seasonally with day length, temperature, water availability and leaf development. Deciduous plants may lose leaves during unfavorable seasons, reducing water loss or avoiding cold damage. Evergreen plants retain leaves, but their photosynthetic activity can still slow during cold or dry periods.

At the ecosystem scale, satellite observations can estimate vegetation greenness and seasonal patterns. Greenness is useful but is not a direct measurement of carbon fixation in every situation. Species composition, leaf structure, stress and background surfaces can influence remote-sensing signals.

10. Experimental Methods and Data Interpretation

10.1 Measuring photosynthesis

Photosynthetic rate can be estimated in several ways. One approach measures oxygen production by aquatic plants under controlled illumination. Another measures carbon dioxide uptake using gas sensors or infrared gas analyzers. Researchers may also measure changes in dissolved oxygen, biomass accumulation or isotope incorporation.

A good experiment defines the response variable, identifies the independent variable and controls relevant conditions. For example, when testing the effect of light intensity, carbon dioxide concentration, temperature, plant size and measurement time should be kept as constant as practical. Replicate measurements help estimate variability.

10.2 Example investigation: light intensity

An aquatic plant can be placed in water containing a controlled supply of dissolved carbon dioxide. A lamp is positioned at different distances, and oxygen production is measured over equal time intervals. The distance from the lamp is changed to alter light intensity. The plant should be allowed to acclimate after each adjustment before measurements are recorded.

Potential sources of error include changes in water temperature caused by the lamp, bubbles sticking to the plant, inconsistent plant surface area and variation in dissolved carbon dioxide. A heat filter or water bath can reduce temperature changes. Repeating each distance measurement and calculating a mean improves reliability.

10.3 Interpreting a response curve

If photosynthetic rate is plotted against light intensity, the curve often rises at low intensity and then approaches a plateau. The initial rising region indicates that light is limiting. The plateau indicates that another factor limits the rate or that the photosynthetic machinery has reached its maximum under the tested conditions.

If the experiment is repeated at higher carbon dioxide concentration, the plateau may occur at a higher photosynthetic rate. This suggests that carbon dioxide was limiting in the original high-light region. Such conclusions require careful controls because changing one environmental variable can unintentionally alter others.

10.4 Correlation and causation

A correlation between light intensity and plant growth does not, by itself, prove that light caused the observed difference. Plants receiving more light may also experience different temperatures or water loss. Controlled experiments help isolate causal relationships. Observational studies are valuable for studying natural systems, but they require careful consideration of confounding variables.

11. Worked Examples and Concept Connections

11.1 Following carbon through photosynthesis

Consider a carbon dioxide molecule entering a leaf through an open stoma. It diffuses through internal air spaces and dissolves in moist cell surfaces. In a chloroplast, carbon dioxide can be fixed by RuBisCO when it reacts with RuBP. The carbon becomes part of 3-PGA, is reduced to G3P using ATP and NADPH, and may eventually become part of sucrose, starch or cellulose.

This sequence connects gas exchange, chloroplast structure, the Calvin cycle and plant growth. It also shows why a limitation at one stage can influence the whole process. If stomata close, less carbon dioxide enters; carbon fixation may decline even if light is abundant.

11.2 Following energy through the chloroplast

A photon absorbed by a pigment excites an electron in a photosystem. Electron transfer contributes to a proton gradient across the thylakoid membrane. ATP synthase uses the gradient to make ATP, while electrons from photosystem I help reduce NADP+ to NADPH. ATP and NADPH are then used in the stroma to support carbon fixation.

Energy is transformed at each stage rather than created. Light energy becomes electrochemical potential energy, then chemical energy in ATP and NADPH, and ultimately chemical energy stored in organic molecules.

11.3 Why a plant can wilt in bright sunlight

Bright sunlight can increase the potential rate of photosynthesis, but it can also increase leaf temperature and evaporative demand. If roots cannot replace water as quickly as it is lost, leaf water potential declines and cells lose turgor. Guard cells may close stomata, limiting further water loss but also restricting carbon dioxide uptake. The result can be wilting and reduced photosynthesis despite abundant light.

11.4 Distinguishing similar terms

Photosynthesis is the overall process that converts light energy into chemical energy stored in organic molecules. Photorespiration is a process initiated when RuBisCO reacts with oxygen instead of carbon dioxide. Cellular respiration releases energy from organic molecules to produce ATP. Transpiration is the loss of water vapor from plant surfaces, mainly through stomata. These processes interact, but they are not interchangeable.

12. Glossary

ATP — adenosine triphosphate, a molecule that transfers usable chemical energy in cells.

Calvin cycle — a series of reactions in the chloroplast stroma that fixes carbon dioxide into carbohydrate precursors using ATP and NADPH.

Chlorophyll — a green photosynthetic pigment that absorbs light energy.

Chloroplast — an organelle where oxygenic photosynthesis occurs in plants and algae.

Chemiosmosis — ATP production powered by the movement of protons down an electrochemical gradient through ATP synthase.

Electron transport chain — a sequence of carriers that transfer electrons and help establish a proton gradient.

G3P — glyceraldehyde-3-phosphate, a three-carbon product of the Calvin cycle.

Granum — a stack of thylakoids inside a chloroplast; plural: grana.

Gross primary production (GPP) — total rate of energy capture or carbon fixation by producers.

Limiting factor — a variable that restricts the rate of a process when it is in insufficient supply.

NADPH — an electron carrier that supplies reducing power for carbon fixation.

Net primary production (NPP) — gross primary production minus producer respiration.

Photorespiration — a pathway initiated when RuBisCO adds oxygen to RuBP.

Phloem — vascular tissue that transports sugars and other organic substances.

RuBisCO — the enzyme that catalyzes carbon dioxide fixation in the Calvin cycle and can also react with oxygen.

RuBP — ribulose-1,5-bisphosphate, the five-carbon carbon-acceptor molecule in the Calvin cycle.

Stoma — a pore in the plant epidermis regulated by guard cells; plural: stomata.

Stroma — the fluid-filled region inside a chloroplast surrounding the thylakoids.

Thylakoid — a flattened membrane sac inside a chloroplast where light-dependent reactions occur.

Transpiration — loss of water vapor from a plant, mainly through stomata.

Xylem — vascular tissue that transports water and mineral ions.

13. Review Questions for Retrieval and Assessment

Recall questions

1. In which chloroplast compartment do the light-dependent reactions occur?

2. In which chloroplast compartment does the Calvin cycle occur?

3. Which molecule supplies replacement electrons to photosystem II?

4. What are the two major products supplied by the light-dependent reactions to the Calvin cycle?

5. Which enzyme catalyzes the attachment of carbon dioxide to RuBP?

6. What is the net carbohydrate product that can leave the Calvin cycle?

7. Which gas is released when water is split during oxygenic photosynthesis?

8. Which plant tissue transports water from roots toward leaves?

9. Which plant tissue transports sugars from sources to sinks?

10. What is the relationship between GPP, respiration and NPP?

Explanation and comparison questions

11. Explain why leaves usually appear green even though chlorophyll is essential for photosynthesis.

12. Explain how a proton gradient across the thylakoid membrane contributes to ATP production.

13. Why does closing stomata conserve water but potentially reduce photosynthesis?

14. Compare the location and timing strategies used by C4 and CAM plants.

15. Explain why photorespiration can increase under hot, dry conditions.

16. Distinguish photosynthesis from cellular respiration in terms of energy transformation and gas exchange.

17. Explain how water moves upward through xylem in a tall plant.

18. Why can a photosynthesis-versus-light-intensity curve reach a plateau?

19. Explain why a plant can respire during daylight even while photosynthesis is occurring.

20. Describe how carbon fixed in a leaf can eventually become part of a root or fruit.

Application and scenario questions

21. A plant is moved from dim light to brighter light, and its oxygen production rises before leveling off. Explain the pattern.

22. A drought-stressed plant closes its stomata at midday. Predict the effect on internal carbon dioxide and carbon fixation.

23. A researcher increases carbon dioxide concentration while keeping light intensity high. Explain why the photosynthetic rate may increase and then plateau.

24. A plant has yellowing older leaves. Which mobile nutrient deficiency might be considered, and why is the symptom not conclusive on its own?

25. A succulent opens its stomata mainly at night. Identify the photosynthetic adaptation and explain its water-saving advantage.

26. A leaf receives intense light but is also exposed to very high temperatures. Explain why photosynthesis might decline.

27. An experiment finds that a lamp increases oxygen production but also warms the water. Identify the confounding variable and propose a control.

28. A forest has high gross primary production but also high plant respiration. Explain how its net primary production can be lower than its GPP.

29. A query asks, 'How do plants make food when it is hot and dry?' Identify at least three concepts from this document needed for a complete answer.

30. A student claims that the Calvin cycle happens in the thylakoid membrane because it uses ATP made there. Correct the claim and explain the connection between the two stages.

Answer key

1. Thylakoid membranes. 2. Stroma. 3. Water. 4. ATP and NADPH. 5. RuBisCO. 6. G3P. 7. Oxygen. 8. Xylem. 9. Phloem. 10. NPP = GPP - producer respiration.

11. Chlorophyll absorbs blue and red light more strongly and reflects or transmits more green light. 12. Protons flow through ATP synthase down their electrochemical gradient, powering ATP formation. 13. Closing stomata reduces water-vapor loss but also reduces carbon dioxide entry. 14. C4 separates initial fixation and the Calvin cycle spatially; CAM separates them temporally, usually taking up CO2 at night. 15. Stomatal closure lowers internal CO2 and makes oxygenation by RuBisCO more likely. 16. Photosynthesis stores light energy in organic molecules; respiration releases energy from organic molecules to make ATP. 17. Transpiration creates tension; cohesion and adhesion help maintain the water column. 18. Another factor becomes limiting or the photosynthetic machinery reaches its capacity under the conditions. 19. Respiration occurs continuously in living cells. 20. Sugars can be transported through phloem from source leaves to sink roots or fruits.

21. More photons initially drive light reactions; at higher light another factor becomes limiting. 22. Internal CO2 and carbon fixation are likely to decline. 23. More CO2 supports carbon fixation until another factor limits the rate. 24. Nitrogen deficiency may cause older leaves to yellow because nitrogen is mobile; other stresses can cause similar symptoms. 25. CAM; nighttime uptake reduces water loss because conditions are cooler and often more humid. 26. Enzyme performance, stomatal opening and membrane function can be impaired at excessive temperatures. 27. Temperature; use a water bath or heat filter and monitor temperature. 28. NPP is GPP minus respiration. 29. Stomatal closure, water stress, temperature, photorespiration and C4 or CAM adaptations. 30. The Calvin cycle occurs in the stroma; it uses ATP and NADPH produced by the light-dependent reactions in the thylakoid membranes.